

CLAIMS

1. A computer-implemented machine reasoning governance system comprising a commitment-control layer configured to:

- (a) receive or construct a candidate commitment comprising one or more of: a candidate output, candidate determination, candidate action, candidate interpretation, candidate state transition, structured conceptual state, structured state proxy, epistemic state proxy, or decision context;
- (b) evaluate whether the candidate commitment satisfies applicable structural admissibility conditions;
- (c) control whether the candidate commitment is released to output, delivery, execution, reliance, or other consequence, permitting release only when the applicable admissibility conditions are satisfied; and
- (d) otherwise enforce a governed non-commitment outcome comprising either an open non-commitment outcome that preserves a continuation path, or a closed non-commitment outcome that closes the particular candidate commitment;

wherein the commitment-control layer is deployable in full-stack operation using structured conceptual state produced by a reasoning-construction layer, or in reduced runtime operation without requiring deployment of the reasoning-construction layer; and

wherein satisfaction of the applicable admissibility conditions is not determined solely by a probabilistic confidence score.

2. The system of claim 1, wherein, in reduced runtime operation, the commitment-control layer comprises a runtime governance module interposed between an application and a model interface, the model interface comprising a language generation model, large language model, transformer-based model, statistical model, agent, retrieval system, application module, or other output-generating system.

3. The system of claim 2, wherein the runtime governance module intercepts a candidate output before delivery, commitment, execution, downstream reliance, or transmission to an output destination, the output destination comprising one or more of: a user interface, an application interface, a downstream system, an automated workflow, a robotic system, a medical, legal, financial, safety, or operational decision system, another model, an agent, an execution engine, or a public communication channel.

4. The system of claim 2, wherein the model interface operates as a non-authoritative expression layer or candidate-output generator, and does not determine admissibility of commitment, wherein commitment authority is retained by the commitment-control layer.

5. The system of claim 1, wherein, when the candidate commitment comprises multiple candidate interpretations, the commitment-control layer maintains the candidate interpretations concurrently as epistemic states, and permits commitment to one candidate interpretation only when the applicable admissibility conditions eliminate or resolve the

competing candidate interpretations, commitment being prohibited while more than one candidate interpretation remains admissible or no candidate interpretation is admissible.

6. The system of claim 1, wherein the open non-commitment outcome comprises one or more of: constrained clarification, containment, revision, escalation, delayed output, withholding, revalidation or re-attestation requirement, or maintained unresolved state; and wherein the closed non-commitment outcome comprises refusal of the particular candidate commitment as a governed terminal disposition.

7. The system of claim 1, wherein the system further comprises the reasoning-construction layer, the reasoning-construction layer comprising one or more of: concept kernels; reflective semantic topology; a constraint-based reasoning engine; a reflective comparator; expression-layer decoupling; roles; domains; spans; compositional primitives; primitive preconditions; primitive update logic; primitive collapse conditions; interpretability evaluation; parallel interpretation maintenance; deterministic processing cycles; or Persistent Existence Frame state.

8. The system of claim 7, wherein the reasoning-construction layer comprises Persistent Existence Frame state, wherein epistemic states, including unresolved epistemic states, persist across turns, sessions, governed system instances, application contexts, or suspensions of active processing, and wherein the Persistent Existence Frame state remains instantiated or durably represented across suspension of active processing, retains identity and unresolved state, and is restored without the suspension being treated as epistemic collapse or as forcing resolution.

9. The system of claim 2, wherein, in reduced runtime operation, the runtime governance module constructs or receives a structured state proxy or epistemic state proxy for admissibility evaluation, the proxy being constructed by one or more of: (a) parsing a candidate model output to identify candidate entities, referents, bindings, claims, or determinations; (b) extracting slots, tags, or annotations supplied by an application; (c) receiving partial structure from a retrieval system, policy engine, or authority registry; (d) mapping model output tokens or segments to provisional role, domain, or span representations; or (e) receiving unresolved-ambiguity, evidence, authority, or consequence metadata from an upstream component; and wherein admissibility evaluation is performed on the proxy according to the same admissibility conditions applicable in full-stack operation, proxy construction not constituting commitment and not delegating admissibility authority to the model interface.

10. The system of claim 2, wherein the runtime governance module evaluates admissibility of an input before the model interface is invoked, and blocks, resolves, or governs an inadmissible input before the input reaches the model interface, whereby an inadmissible input is prevented from causing generation of a candidate output.

11. The system of claim 2, wherein the runtime governance module determines whether an input is resolvable deterministically from maintained epistemic state, persistent state, or governance state, and, when the input is so resolvable, produces a response or disposition from the maintained state without invoking the model interface, the response or disposition being deterministically reconstructible from the maintained state.

12. The system of claim 1, wherein the applicable structural admissibility conditions comprise one or more of: orientation source currency; counterparty status; sanctions or trade-restriction

status; credential validity; licence validity; evidence freshness or staleness; authority validation; policy compliance; consequence grade; or jurisdictional or institutional rule; and wherein one or more of the admissibility conditions are evaluated using orientation information received from, queried from, polled from, subscribed to, or associated with an external or institutional orientation source comprising one or more of: a regulatory registry, sanctions list, trade-status system, supplier registry, authority registry, credential system, licence registry, HR system, compliance system, policy register, evidence repository, source-trust registry, or operational state system.

13. The system of claim 12, wherein the commitment-control layer evaluates freshness of an orientation condition by one or more of: expiry, time-to-live, revalidation interval, source version, registry version, effective date, supersession status, or jurisdictional change; and wherein a candidate commitment is treated as inadmissible, unresolved, or requiring revalidation when an orientation condition, evidence item, authority state, policy state, source state, counterparty status, supplier status, credential status, licence status, or registry status is stale, expired, superseded, inconsistent, unavailable, or not revalidated within a predetermined interval, notwithstanding that no revocation event, notification, or policy update has been received.

14. The system of claim 13, wherein, after an admissibility-relevant change, expiry, staleness condition, or unresolved orientation condition is detected or inferred, each further execution, release, approval, recommendation, delivery, or downstream reliance is treated as a new candidate commitment requiring admissibility evaluation under the then-current admissibility conditions, and wherein a stale, missing, superseded, inconsistent, or unrevalidated orientation condition does not constitute permission to continue.

15. The system of claim 1, further comprising a governance ledger configured to record a governance event associated with evaluation of a candidate commitment, the governance event including at least the governance outcome and one or more failed or satisfied admissibility conditions, wherein governance records are protected by a tamper-evident integrity mechanism comprising one or more of: a message authentication code, a digital signature, a state hash, a hash chain, an append-only structure, a signed receipt, or other cryptographic authentication or integrity mechanism, and wherein the governance ledger enables reconstruction, replay, audit, or verification of an admissibility determination.

16. The system of claim 1, wherein a single commitment-control layer retains commitment authority over candidate commitments originating from multiple governed system instances, sessions, model interfaces, applications, agents, or workflows, and applies the applicable admissibility conditions consistently across originating systems such that a candidate commitment that is inadmissible when originating from one governed system instance, session, model interface, application, agent, or workflow cannot obtain commitment, release, or consequence by being generated, resubmitted, or rerouted through a different governed system instance, session, model interface, application, agent, or workflow.

17. A computer-implemented method performed by a runtime governance module interposed between an application and a model interface, the method comprising:

- (a) receiving or constructing a candidate commitment comprising one or more of: a candidate output received from the model interface, a candidate determination, a

candidate action, a candidate interpretation, a candidate state transition, a structured state proxy, an epistemic state proxy, or a decision context;

- (b) evaluating whether the candidate commitment satisfies applicable structural admissibility conditions;
- (c) controlling whether the candidate commitment is released to output, delivery, execution, reliance, or other consequence, release being permitted only when the applicable admissibility conditions are satisfied; and
- (d) otherwise enforcing a governed non-commitment outcome comprising either an open non-commitment outcome that preserves a continuation path, or a closed non-commitment outcome that closes the particular candidate commitment;

wherein satisfaction of the applicable admissibility conditions is not determined solely by a probabilistic confidence score.

18. The method of claim 17, further comprising receiving state, constraints, metadata, orientation information, or admissibility information from one or more of: a reasoning-construction layer; a compositional primitive engine; an Aurora Block; a Persistent Existence Frame providing active-state, echo-state, potential-state, unresolved-state, reconstruction, or projection metadata; a retrieval system providing evidence metadata; a source registry or source-trust system; an authority registry; a policy engine; an application layer providing consequence grade, user role, or action context; a model interface; or an orientation source providing operational, regulatory, institutional, counterparty, sanctions, trade-status, credential, licence, or jurisdictional information; and evaluating admissibility of the candidate commitment using any combination of the received inputs.

19. The method of claim 17, wherein a completed governance event, including its governance outcome and recorded disposition, is final with respect to the particular candidate commitment evaluated, such that the recorded disposition is not retroactively altered, widened, softened, or reversed by a subsequently received input, turn, or state change; and wherein a subsequently received input, changed condition, newly satisfied admissibility condition, or revalidated orientation condition may initiate a new, separately identified candidate commitment evaluated under the then-current admissibility conditions, without modifying the recorded disposition of the prior governance event.

20. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause the one or more processors to:

- (a) receive or construct a candidate commitment comprising one or more of: a candidate output received from a model interface, a candidate determination, a candidate action, a candidate interpretation, a candidate state transition, a structured state proxy, an epistemic state proxy, or a decision context;
- (b) evaluate whether the candidate commitment satisfies applicable structural admissibility conditions;

- (c) control whether the candidate commitment is released to output, delivery, execution, reliance, or other consequence, release being permitted only when the applicable admissibility conditions are satisfied; and
- (d) otherwise enforce a governed non-commitment outcome comprising either an open non-commitment outcome that preserves a continuation path, or a closed non-commitment outcome that closes the particular candidate commitment;

wherein satisfaction of the applicable admissibility conditions is not determined solely by a probabilistic confidence score.